WORKED SOLUTION

STEP II 1993, Question 16

Probability and Statistics • Poisson Processes

STEP II 1993, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q16

The key idea

There are two independent sources of randomness here: the buses, which give a geometric waiting time, and the passengers, who form a Poisson process. Condition on when the first bus comes. Once you know that, the number waiting is just a Poisson count over a known length of time. Part (iii) then uses the law of total probability, and the real question is how to evaluate the resulting infinite sum to two significant figures with only tables and a calculator.

Solution

Measure time in minutes after 8:00. Let T be the time at which the first bus arrives, and let X_t be the number of passengers who arrive in $(0, t]$. The passenger process is Poisson with rate 4 per minute, so $X_t \sim \text{Po}(4t)$. Buses can come only at $t = 1, 2, 3, \ldots$ ("just after eight" means the bus that may come at 8:00 has already been and gone), and they are independent of the passengers.

Part (i)

The first bus arrives at n minutes past 8 exactly when no bus comes at minutes $1, 2, \ldots, n-1$ and a bus does come at minute n. These events are independent, so

$$\text{P}(T = n) = \left(\tfrac{3}{4}\right)^{n-1} \tfrac{1}{4}, \qquad n = 1, 2, 3, \ldots$$

So T is geometric with parameter $\frac{1}{4}$. (As a check, these probabilities sum to $\frac{1}{4}/(1 - \frac{3}{4}) = 1$.)

Part (ii)

The people waiting at 8:05 are exactly those who arrived in $(0, 5]$. Nobody has left, because no bus has come before 8:05. So we need the distribution of X_5 given $T = 5$.

Ask yourself Does knowing that the first bus came at 8:05 change the distribution of the number of passengers?

No. The event $T = 5$ depends only on the buses, and the buses are independent of the passengers. So conditioning on $T = 5$ leaves the distribution of X_5 unchanged, and $X_5 \sim \text{Po}(4 \times 5) = \text{Po}(20)$. Hence

$$\text{P}(m \text{ people waiting} \mid T = 5) = \frac{\text{e}^{-20}\, 20^m}{m!}, \qquad m = 0, 1, 2, \ldots$$

Part (iii)

The bus fails to take everyone exactly when more than 25 people are waiting, that is, when $X_T \geq 26$. Let F be this event. Conditioning on T and using independence as in (ii),

$$\text{P}(F) = \sum_{n=1}^{\infty} \text{P}(T = n)\, \text{P}(X_n \geq 26) = \sum_{n=1}^{\infty} \left(\tfrac{3}{4}\right)^{n-1} \tfrac{1}{4}\, \text{P}(\text{Po}(4n) \geq 26).$$

We now need a plan that turns this infinite sum into a finite amount of work, with an error well below 0.005. We want two significant figures, and the answer will turn out to be of order 0.1.

Step 1: work with the complement, so the tail is easy. Write $w_n = \left(\tfrac{3}{4}\right)^{n-1} \tfrac{1}{4}$ and $q_n = \text{P}(\text{Po}(4n) \leq 25)$. Since $\sum w_n = 1$,

$$\text{P}(F) = 1 - \sum_{n=1}^{\infty} w_n\, q_n.$$

This form is better because q_n is tiny when n is large. For $n \geq 11$ the mean $4n \geq 44$ is far above 25. Since q_n decreases as n increases (more time means more passengers),

$$\sum_{n \geq 11} w_n q_n \leq q_{11} \sum_{n \geq 11} w_n = q_{11} \left(\tfrac{3}{4}\right)^{10}.$$

A normal approximation gives $q_{11} \approx \Phi((25.5 - 44)/\sqrt{44}) = \Phi(-2.79) \approx 0.003$, so this tail is below 0.0002 and can be dropped. That leaves ten terms.

Step 2: the first few terms. For $n = 1$ and $n = 2$ (means 4 and 8), getting 26 or more passengers is so unlikely that $q_1 = q_2 = 1$ to well beyond the accuracy needed, so these contribute $w_1 + w_2 = \tfrac{1}{4} + \tfrac{3}{16}$.

Step 3: the middle terms from tables. For $n = 3, 4, 5$ (means 12, 16, 20), read $q_n = \text{P}(\text{Po}(4n) \leq 25)$ directly from cumulative Poisson tables, which normally cover means up to 20. If the tables also list means 24 and 28, use them for $n = 6, 7$ as well.

Step 4: larger means by the normal approximation. For the remaining n up to 10 (means up to 40), use $\mathrm{Po}(\mu) \approx \mathrm{N}(\mu, \mu)$ with a continuity correction:

$$q_n \approx \Phi\left(\frac{25.5 - 4n}{\sqrt{4n}}\right),$$

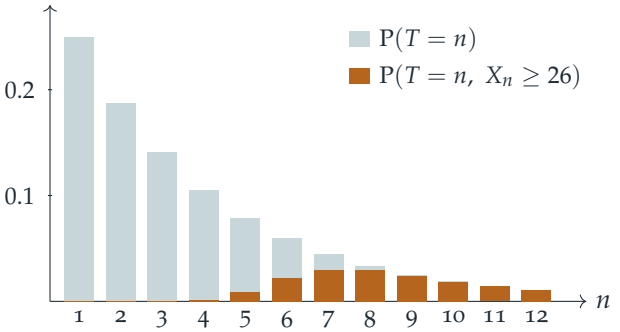
reading Φ from normal tables. These means are large enough for the approximation to be good. More to the point, the weights w_n are already small ($w_8 \approx 0.033$, $w_{10} \approx 0.019$), so an error of a few thousandths in q_n changes $\mathrm{P}(F)$ by less than 0.001.

Step 5: assemble. Compute $w_1, \ldots, w_{10}$ on the calculator and form

$$\mathrm{P}(F) \approx 1 - \sum_{n=1}^{10} w_n q_n,$$

then round to two significant figures. Each neglected or approximated piece is at most about 0.001, so the rounding is safe unless the total lands very close to a rounding boundary. If it does, the method shows exactly which terms to refine.

The question doesn't ask us to carry this out, but doing so gives $\mathrm{P}(F) \approx 0.19$. The chart below shows where the probability comes from: most of it comes from first buses at 8:06 to 8:10, when the expected queue of 24 to 40 is close to or above capacity.



Checks

- In (i), the probabilities sum to 1, and $\mathrm{P}(T = 1) = \frac{1}{4}$ as it should.
- In (ii), the mean number waiting is 20, which is four per minute for five minutes.
- In (iii), the exact sum (computed by machine) is 0.1904. A Monte Carlo simulation of 2×10^5 mornings gives 0.189. The table-and-normal method above gives 0.1912. All three round to 0.19.

Common slips

- Getting the boundary wrong by one. The bus fails only if 26 or more people are waiting, so the probability we need is $1 - \mathrm{P}(\mathrm{Po}(4n) \leq 25)$, not $1 - \mathrm{P}(\mathrm{Po}(4n) \leq 24)$.

- Treating the queue length as $\mathrm{Po}(4\,\mathrm{E}[T]) = \mathrm{Po}(16)$. The waiting time is random, so you must average the Poisson probabilities over T, not plug in the average time.
- Leaving the sum infinite without saying why it can be cut off. You must show that the neglected tail is too small to affect the second significant figure.

Beyond STEP

The number waiting, X_T, is a Poisson count with a random mean: a *mixture* of Poisson distributions. Because T is geometric, X_T turns out to have a neat closed-form generating function, and conditioning arguments like this one are the basis of queueing theory and of the compound and mixed distributions met in a first probability course. The question of how to cut off a sum and bound the error is the start of numerical analysis. Grimmett and Welsh, *Probability: An Introduction*, covers Poisson processes and generating functions at the right level.

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